11/04/2025

CODE-A

EDITION 2025 EDITION

# Aakash

Medical | IIT-JEE | Foundations

***Corporate Office*: AESL, 3rd Floor, Incuspaze Campus-2, Plot-13, Sector-18, Udyog Vihar, Gurugram, Haryana-122015**

720

# AIM – 720

## (*Advanced* INTENSIVE *Mastery for* 720)

**MM : 720**

## CST-3

**Time : 180 Mins.**

### Answers

| | | | | |
|---|---|---|---|---|
| 1. (1) | 37. (2) | 73. (3) | 109. (1) | 145. (4) |
| 2. (3) | 38. (2) | 74. (1) | 110. (4) | 146. (3) |
| 3. (2) | 39. (3) | 75. (3) | 111. (3) | 147. (2) |
| 4. (4) | 40. (2) | 76. (4) | 112. (1) | 148. (4) |
| 5. (3) | 41. (3) | 77. (2) | 113. (3) | 149. (2) |
| 6. (3) | 42. (1) | 78. (2) | 114. (2) | 150. (4) |
| 7. (2) | 43. (4) | 79. (3) | 115. (4) | 151. (2) |
| 8. (3) | 44. (1) | 80. (2) | 116. (3) | 152. (2) |
| 9. (3) | 45. (4) | 81. (3) | 117. (3) | 153. (4) |
| 10. (2) | 46. (3) | 82. (3) | 118. (4) | 154. (4) |
| 11. (2) | 47. (2) | 83. (3) | 119. (1) | 155. (3) |
| 12. (1) | 48. (2) | 84. (4) | 120. (4) | 156. (1) |
| 13. (4) | 49. (1) | 85. (2) | 121. (3) | 157. (2) |
| 14. (4) | 50. (3) | 86. (2) | 122. (2) | 158. (4) |
| 15. (4) | 51. (2) | 87. (3) | 123. (4) | 159. (1) |
| 16. (4) | 52. (4) | 88. (4) | 124. (3) | 160. (4) |
| 17. (4) | 53. (4) | 89. (2) | 125. (2) | 161. (2) |
| 18. (4) | 54. (1) | 90. (2) | 126. (3) | 162. (2) |
| 19. (2) | 55. (2) | 91. (1) | 127. (1) | 163. (3) |
| 20. (4) | 56. (3) | 92. (3) | 128. (1) | 164. (4) |
| 21. (3) | 57. (1) | 93. (2) | 129. (3) | 165. (4) |
| 22. (3) | 58. (2) | 94. (4) | 130. (3) | 166. (1) |
| 23. (1) | 59. (3) | 95. (2) | 131. (3) | 167. (4) |
| 24. (1) | 60. (3) | 96. (1) | 132. (4) | 168. (3) |
| 25. (4) | 61. (3) | 97. (4) | 133. (4) | 169. (2) |
| 26. (3) | 62. (1) | 98. (2) | 134. (4) | 170. (2) |
| 27. (4) | 63. (2) | 99. (2) | 135. (4) | 171. (4) |
| 28. (2) | 64. (1) | 100. (2) | 136. (2) | 172. (3) |
| 29. (3) | 65. (1) | 101. (1) | 137. (4) | 173. (2) |
| 30. (1) | 66. (3) | 102. (2) | 138. (1) | 174. (4) |
| 31. (2) | 67. (2) | 103. (1) | 139. (4) | 175. (4) |
| 32. (4) | 68. (3) | 104. (4) | 140. (2) | 176. (4) |
| 33. (3) | 69. (2) | 105. (1) | 141. (3) | 177. (3) |
| 34. (2) | 70. (4) | 106. (1) | 142. (1) | 178. (3) |
| 35. (2) | 71. (2) | 107. (2) | 143. (3) | 179. (4) |
| 36. (3) | 72. (4) | 108. (4) | 144. (4) | 180. (4) |

## Hints and Solutions

### PHYSICS

**(1)** **Answer :** (1)
**Solution:**
Wavelength of radiation emitted is given as
$\frac{1}{\lambda} = Rz^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$
$z$ = 1, for hydrogen atom
$\frac{1}{\lambda} = R\left[\frac{1}{9} - \frac{1}{16}\right]$
$\frac{1}{\lambda} = R\left[\frac{16-9}{9\times16}\right]$
$\frac{1}{\lambda} = R\left[\frac{7}{144}\right]$
$\lambda = \frac{144}{7R}$

**(2)** **Answer :** (3)
**Solution:**
$Fe^{56}$ nucleus is more stable than $U^{235}$ because binding energy per nucleon is more in $Fe^{56}$ than $U^{235}$.

**(3)** **Answer :** (2)
**Solution:**
∴ Given truth table is for NOR gate
$Y = \overline{A+B} = \overline{A}\cdot\overline{B}$

**(4)** **Answer :** (4)
**Solution:**
Dynamic resistance of the reverse biased diode
$R = \frac{\Delta V}{\Delta I}$
$R = \frac{5}{2\times10^{-6}}$
$R = 2.5 \times 10^6\ \Omega$

**(5)** **Answer :** (3)
**Solution:**
For balanced bridge,
$\frac{3}{R} = \frac{25}{100-25} = \frac{1}{3}$
$R = 9\ \Omega$

**(6)** **Answer :** (3)
**Solution:**
According to Einstein's photoelectric equation,
$(KE)_{max} = hf - \phi$
$(KE)_{max} = 6.1 - 4.2 = 1.9$ eV
$= 1.9 \times 1.6 \times 10^{-19}$ J
$= 3.04 \times 10^{-19}$ J

**(7)** **Answer :** (2)
**Solution:**
$\bar{l} = \frac{3.29+3.28+3.29+3.31+3.28}{5}$
$\bar{l} = 3.29$ cm
Most accurate length of cylinder will be the mean length.

**(8)** **Answer :** (3)
**Solution:**
$V_{avg} = \frac{\text{Total distance}}{\text{Total time}} = \frac{2v_1v_2}{v_1+v_2}$

$V_{\text{avg}} = \frac{2\times5\times10}{15} = \frac{20}{3}$

**(9)** **Answer :** (3)
**Solution:**
$s = \left(\frac{v+u}{2}\right)t$
$= \left(\frac{2+3}{2}\right) \times 2 = 5$ m

**(10)** **Answer :** (2)
**Solution:**
$v_X = v\cos60°$
$= 10 \times \frac{1}{2} = 5$ m/s
In horizontal direction
$p_f = p_i$
$60 \times v_D = 1 \times 10 \times \frac{1}{2} \Rightarrow v_D = \frac{1}{12}\ \text{m/s}$
$T = \frac{2v\sin 60^\circ}{10} = 2 \times 10 \times \frac{\sqrt{3}}{20} = \sqrt{3}$ s
Displacement of Dashrath = $v_D \times T = \frac{1}{12} \times \sqrt{3} = \frac{1}{4\sqrt{3}}\ \text{m}$

**(11)** **Answer :** (2)
**Solution:**
$\frac{1}{2} \times 6 \times 10^{-3} \times F_0 = 0.1(20+10)$
$3 \times 10^{-3} F_0 = 0.1 \times 30$
$F_0 = 10^3$ N

**(12)** **Answer :** (1)
**Solution:**
$m_{\text{Raunak}} = \frac{m_{\text{Arun}}}{2}$
$(K.E)_{\text{Arun}} = \frac{1}{3}(K.E.)_{\text{Raunak}}$
$\frac{1}{2} \times m_{\text{Arun}} \times v'^2 = \frac{1}{3} \times \frac{1}{2} \times \frac{m_{\text{Arun}}}{2} \times v^2$
$v'^2 = \frac{1}{6}v^2$
$v' = \frac{v}{\sqrt{6}}$

**(13)** **Answer :** (4)
**Solution:**
$P = \frac{mgh}{t}$
$m = \frac{Pt}{gh}$
$m = \frac{100\times10^3\times2}{10\times20}$
$m = 10^3$ kg

**(14)** **Answer :** (4)
**Solution:**
The magnitude of difference between true value or actual value and measured value in a experiment is called absolute error.

**(15)** **Answer :** (4)
**Solution:**
During projectile motion, acceleration remain constant and equal to $g$ (downward).

**(16)** **Answer :** (4)
**Solution:**
For C.O.M.
$X_{\text{CM}} = \frac{m\times0+2m\times a+4m\times\frac{a}{2}}{m+2m+4m} = \frac{4ma}{7m} = \frac{4a}{7}$
$Y_{\text{CM}} = \frac{m\times0+2m\times0+4m\times\frac{\sqrt{3}a}{2}}{m+2m+4m}$

$= \frac{2\sqrt{3}ma}{7m} = \frac{2\sqrt{3}a}{7}$

**(17)** **Answer :** (4)
**Solution:**
$E_i = E = -\frac{GMm}{2r}$
$E_f = -\frac{GMm}{4r}$
∴ Energy required = $E_f - E_i = \frac{GMm}{2r} - \frac{GMm}{4r} = \frac{GMm}{4r} = \frac{-E}{2}$

**(18)** **Answer :** (4)
**Solution:**
Mass density of turpentine is less than that of water but optical density of turpentine is higher.

**(19)** **Answer :** (2)
**Solution:**
$\frac{1}{v_0} - \frac{1}{u_0} = \frac{1}{f_0}$
$\Rightarrow \frac{1}{v_0} - \frac{1}{-5} = \frac{1}{1.25} \Rightarrow v_0 = \frac{5}{3}$ cm
For eye-piece lens,
$\frac{1}{-25} - \frac{1}{u_e} = \frac{1}{f_e}$
$\Rightarrow \frac{1}{u_e} = \frac{-1}{25} - \frac{1}{12.5} \Rightarrow u_e = -\frac{25}{3}$ cm
$\therefore L = |v_0| + |u_e| = \frac{5}{3} + \frac{25}{3} = 10$ cm

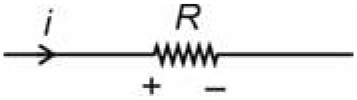
**(20)** **Answer :** (4)
**Solution:**
From the earth's surface,
escape speed $v_e = \sqrt{\frac{2GM_e}{R_e}} = \sqrt{2gR_e}$ is independent of mass of object.
From a height $h$, $v_e = \sqrt{\frac{2GM_e}{R_e+h}} = \sqrt{\frac{2gR_e^2}{R_e+h}}$
Hence escape speed depends on mass of earth and height of projection.

**(21)** **Answer :** (3)
**Solution:**
$\vec{\tau} = \vec{r} \times \vec{F} = rF_{\perp}$ N m
$= 1 \times 6(-\hat{k})$ Nm $= (-6\hat{k})$ N m

**(22)** **Answer :** (3)
**Solution:**
$F = \frac{kq_1q_2}{d^2},\ q_1 + q_2 = Q_2$
We can write, $q_1q_2 = \frac{(q_1+q_2)^2-(q_1-q_2)^2}{4}$
$q_1q_2 = \frac{Q^2-(q_1-q_2)^2}{4}$
For $F_{max}$, we need $q_1q_2$ to be maximum.
$\therefore q_1 - q_2 = 0 \Rightarrow q_1 = q_2$
Hence $q_1 = q_2 = \frac{Q}{2}$

**(23)** **Answer :** (1)
**Solution:**
For a single small drop, we can write
$V = \frac{kq}{r} = 1$ V
For large drop:
$V' = \frac{k \times 1000q}{R}$
From volume conservation
$\frac{4}{3}\pi R^3 = 1000 \times \frac{4}{3}\pi r^3 \Rightarrow R = 10r$
$\therefore V' = \frac{k \times 1000q}{10r} = \frac{kq}{r} \times 100$
$V' = 1 \times 100$ volt

**(24) Answer :** (1)
**Solution:**
Batteries have internal resistance and they dissipate energy due to it.




In a resistor, current flows from higher to lower potential.

**(25) Answer :** (4)
**Solution:**
Applying wheat stone bridge principal:
$\frac{R_1}{40} = \frac{R_2}{100-40} \Rightarrow \frac{R_1}{R_2} = \frac{2}{3}$ ...(i)
When $R_2$ is replaced by $2R_2$:
$\frac{R_1}{x} = \frac{2R_2}{100-x} \Rightarrow \frac{R_1}{R_2} = \frac{2x}{100-x}$
$\frac{2}{3} = \frac{2x}{100-x} \Rightarrow x = 25$ cm

**(26) Answer :** (3)
**Solution:**
Placing a dielectric inside a parallel plate capacitor increases its capacity.

**(27) Answer :** (4)
**Solution:**
By changing the reference of potential at infinity, potential at each point changes but the potential difference between points remains constant. Hence the potential difference between $A$ and $B$ will remain same as 10 V.

**(28) Answer :** (2)
**Solution:**
$a_R = \frac{v^2}{R}$
$R = \frac{mv}{qB}$
$a_R = \left(\frac{qBv}{m}\right)$

**(29) Answer :** (3)
**Solution:**
In uniform magnetic field.
$F = i(\vec{l}_{\text{eff}} \times \vec{B})$
$= IlB$

**(30) Answer :** (1)
**Solution:**
$X_C = \frac{1}{\omega C}$ and $X_L = \omega L$

**(31) Answer :** (2)
**Solution:**
$L_{\text{eff}} = \frac{3\times6}{3+6} + 2$
= 2 + 2 = 4 H

**(32) Answer :** (4)
**Solution:**
$I = \frac{M}{V}$
$= \frac{2.5}{10\times5\times4\times10^{-6}} = \frac{25\times10^5}{25\times8}$
$= \frac{1}{8} \times 10^5 = 1.25 \times 10^4$ A m$^{-1}$

**(33) Answer :** (3)
**Solution:**
$|\varepsilon| = L\frac{dI}{dt}$
$L = \frac{|\varepsilon|dt}{(dI)}$

$= \frac{\text{Volt-second}}{\text{Ampere}}$

**(34) Answer :** (2)

**Solution:**

$i = i_0 \sin \omega t$

$\frac{i_0}{\sqrt{2}} = i_0 \sin \omega t$

$\sin \omega t = \frac{1}{\sqrt{2}}$

$\omega t = \frac{\pi}{4}$

$t = \left(\frac{\pi}{4\omega}\right)$

**(35) Answer :** (2)

**Solution:**

Strain is unitless and dimensionless

$Y = \frac{\text{Stress}}{\text{Strain}}$; [$Y$] = [stress] = [$ML^{-1}T^{-2}$]

**(36) Answer :** (3)

**Solution:**

$x = A \sin(\omega t + \phi_0)$

$f = 25$ Hz, $\phi_0 = \frac{\pi}{4}$

$x = 0.1 \sin(2\pi f t + \phi_0)$

$x = 0.1 \sin\left(50\pi t + \frac{\pi}{4}\right)$

**(37) Answer :** (2)

**Solution:**

Velocity of sound in a gas is

$v = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M}}$

$v \propto \sqrt{T}$

$\propto \frac{1}{\sqrt{M}}$

*v* is independent of change in pressure of gas at constant temperature.

**(38) Answer :** (2)

**Solution:**

UV rays are used to destroy the bacteria and sterilizing the surgical instruments.

**(39) Answer :** (3)

**Solution:**

Two sources are said to be coherent if they emit same colour or vibrate with a constant phase difference.

**(40) Answer :** (2)

**Solution:**

$dw = PdV \Rightarrow dw = (a + bV)dV$

$\Rightarrow dw = adV + bVdV$

$\Rightarrow w = \int dw = \int_2^3 adV + \int_2^3 bVdV$

$\Rightarrow w = a[3-2] + \frac{b}{2}[9-4]$

$= a \times 1 + \frac{5b}{2} = a + \frac{5b}{2}$

**(41) Answer :** (3)

**Solution:**

$AB \rightarrow$ isothermal $\Rightarrow \Delta Q = \Delta W = 700$ J

$BC \rightarrow$ adiabatic $\Rightarrow \Delta Q = 0$

$CA \rightarrow \Delta Q = -100$ J

$\Rightarrow (\Delta Q)_{net} = 700 - 100 = 600$ J

**(42) Answer :** (1)

**Solution:**

Equating the pressure at base,
$\rho_w \times g \times h + \rho_w \times g \times 2 = \rho \times g \times 2 + \rho \times g \times h$
$\Rightarrow \rho_w \times h = \rho \times h \Rightarrow \rho = \rho_w$

**(43) Answer :** (4)
**Solution:**
$\Delta P = \frac{4T}{r} \Rightarrow \Delta P \propto \frac{1}{r}$ as $r \downarrow \Delta P \uparrow$

**(44) Answer :** (1)
**Solution:**
$V_{\text{rms}} = \sqrt{\frac{V_1^2 + V_2^2 + V_3^3 + \cdots V_n^2}{n}}$
$= \sqrt{\frac{1+4+9}{4}} = \sqrt{\frac{7}{2}} = \sqrt{3.5}$ m/s

**(45) Answer :** (4)
**Solution:**
At melting point liquid and solid states coexist.
At boiling point liquid and vapour states coexist.

**CHEMISTRY**

**(46) Answer :** (3)
**Solution:**
Basic strength of hydroxides of lanthanoids decreases down the group.

**(47) Answer :** (2)
**Solution:**
$BaCl_2 + H_2SO_4 \rightarrow BaSO_4 + 2HCl$
Molarity of $BaCl_2$ = $\frac{20.8 \times 1000}{100 \times 208} = 1\ \mathrm{M}$
Molarity of $H_2SO_4$ = $\frac{9.8 \times 100}{100 \times 98} = 1\ \mathrm{M}$
Mole of $BaCl_2$ = 100 × 1 × $10^{-3}$ = 0.1
Mole of $H_2SO_4$ = 50 × $10^{-3}$ = 0.05
Mole of $BaSO_4$ = 0.05 ( $\because$ $H_2SO_4$ = is limiting reagent)

**(48) Answer :** (2)
**Solution:**
$H_2(g) + Br_2(l) \rightarrow 2HBr(g)$
$\Delta_r H = \mathrm{B.E}_{(H_2)} + \mathrm{B.E}_{(Br_2)} - 2 \times \mathrm{B.E}_{(HBr)}$
– 109 = 435 + x – 2 × 368
– 109 = 435 + x – 736
– 109 – 435 + 736 = x
– 544 + 736 = x
x = 192 kJ $mol^{-1}$

**(49) Answer :** (1)
**Solution:**
$\mu = \sqrt{n(n+2)}\ \mathrm{BM}$
$[Fe(CN)_6]^{3-}$ ; n = 1, μ = 1.73 BM
$[FeF_6]^{3-}$ ; n = 5, μ = 5.9 BM
$[Fe(CN)_6]^{4-}$ ; n = 0, μ = 0 BM
$[CoF_6]^{3-}$ ; n = 4, μ = 4.9 BM

**(50) Answer :** (3)
**Solution:**
For free expansion of an ideal gas under adiabatic conditions,
q = 0
and $P_{ext}$ = 0

∴ w = 0

**(51) Answer :** (2)
**Solution:**
The correct order of increasing field strength of ligands is:
$SCN^- < C_2O_4^{2-} < NCS^- < NH_3$

**(52) Answer :** (4)
**Solution:**
$\Delta_a H^\circ\left(V\right) = 515$ kJ mol$^{-1}$

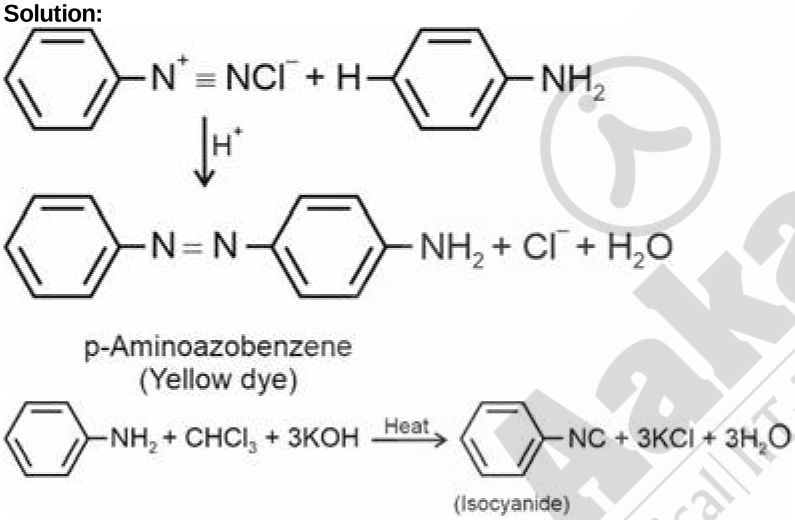
$Cr^{+6} \xrightarrow{H^+} Cr^{3+}$

**(53) Answer :** (4)
**Solution:**

| | Amines | pKb value |
|---|---|---|
| (a) | Methanamine | 3.38 |
| (b) | N-Methylmethanamine | 3.27 |
| (c) | N-N-Dimethylmethanamine | 4.22 |

$pK_b$ Value increases, basicity decreases
Correct order of basicity of (b) > (a) > (c)

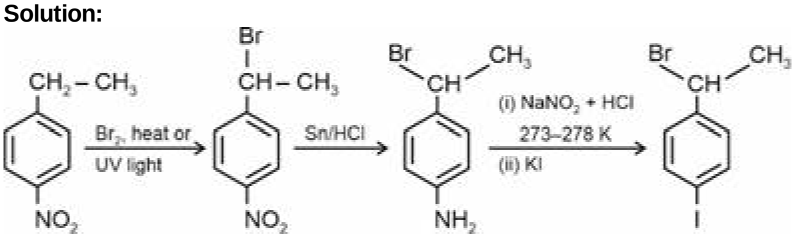
**(54) Answer :** (1)
**Solution:**




**(55) Answer :** (2)
**Solution:**

| Haloalkane | Dipole moment/Debye |
|---|---|
| $CH_3 - F$ | 1.847 |
| $CH_3 - Cl$ | 1.860 |
| $CH_3 - Br$ | 1.830 |
| $CH_3 - I$ | 1.636 |

**(56) Answer :** (3)
**Solution:**




**(57) Answer :** (1)

**Solution:**
Two solutions having same osmotic pressure at a given temperature are called isotonic solutions. When two isotonic solutions are separated by semipermeable membrane no osmosis occurs between them.

**(58) Answer :** (2)
**Solution:**
$\Delta T_f = \frac{K_f \times w_2 \times 1000}{M_2 \times w_1}$
$= \frac{1.86 \times 31 \times 1000}{62 \times 750} = 1.24$
$T_f = -1.24°C$

**(59) Answer :** (3)
**Solution:**
Dipole moment of $NH_3$ is greater than that of $NF_3$. This is because is case of $NH_3$ the orbital dipole due to lone pair is in the same direction as the resultant dipole moment of the N–H bonds.

**(60) Answer :** (3)
**Solution:**
$I_3^-$ , $CO_2$ and $BeCl_2$ are linear.

**(61) Answer :** (3)
**Solution:**
$N_2^+,\ N_2^-,\ O_2^-$ , are paramagnetic species.

**(62) Answer :** (1)
**Solution:**
For zero order reaction
$t_{100\%} = 2t_{50\%} = \frac{a_0}{k}$
First order reaction never reaches completion.

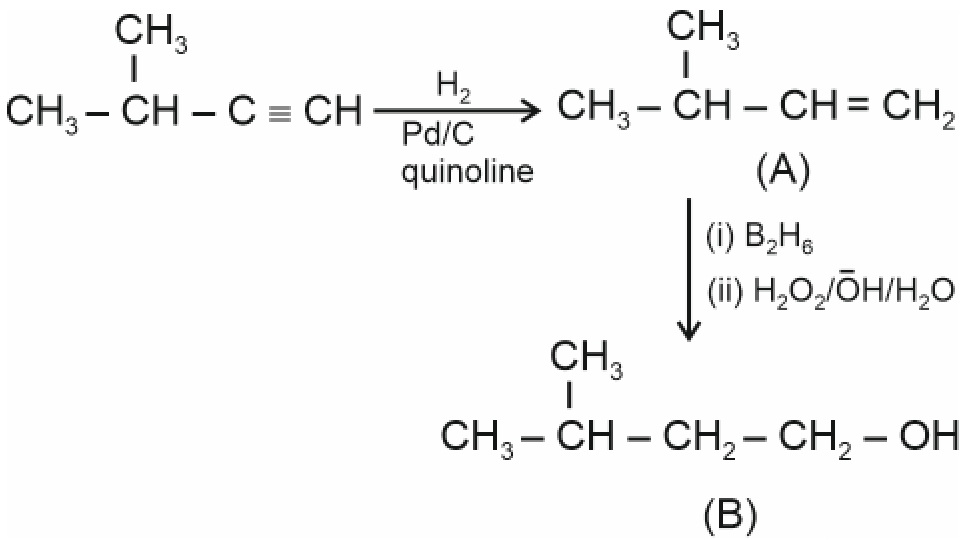
**(63) Answer :** (2)
**Solution:**
$\log \frac{r_2}{r_1} = \frac{Ea}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$
$\log 2 = \frac{Ea}{2.3 \times R}\left[\frac{1}{300} - \frac{1}{330}\right]$
$Ea = 18.96\ kJ\ mol^{-1}$

**(64) Answer :** (1)
**Solution:**
Carbonyl compounds with two α–H atoms give aldol condensation reaction.

**(65) Answer :** (1)
**Solution:**
On adding dilute $H_2SO_4$ to the solid salt, if there is evolution of a colourless and odourless gas with brisk effervescence then carbonate could be present.

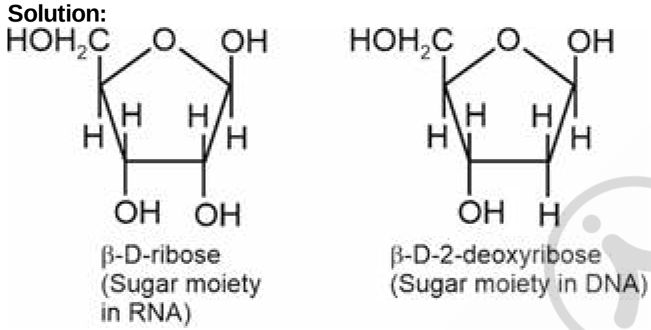
**(66) Answer :** (3)
**Solution:**
$Zn^{2+}$ is white when cold.

**(67) Answer :** (2)
**Solution:**



**(68)** **Answer :** (3)

**Solution:**




**(69)** **Answer :** (2)

**Solution:**

Cathode rays start from cathode and move towards the anode.

**(70)** **Answer :** (4)

**Solution:**

Glucose does not react with $NaHSO_3$

**(71)** **Answer :** (2)

**Solution:**

$$2\,CH_4 + O_2 \xrightarrow[100\text{ atm}]{Cu,\,523K} 2\,CH_3\,OH$$

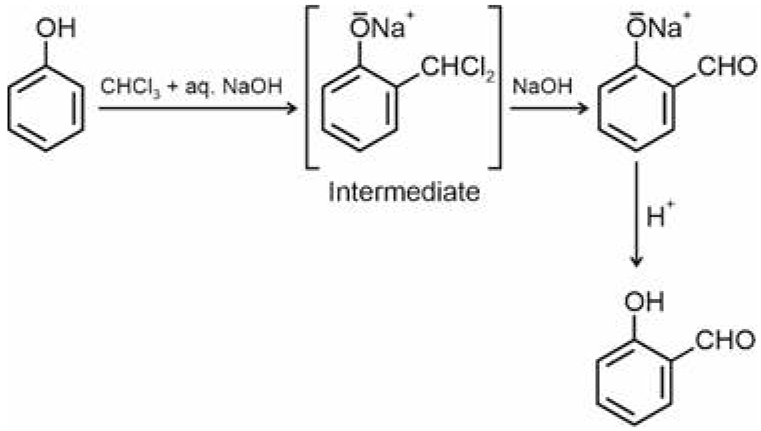
$$CH_4 + H_2O \underset{\Delta}{\overset{Ni}{\rightarrow}} CO + 3H_2$$

$$CH_4 + O_2 \xrightarrow[\Delta]{Mo_2\,O_3} HCHO + H_2O$$

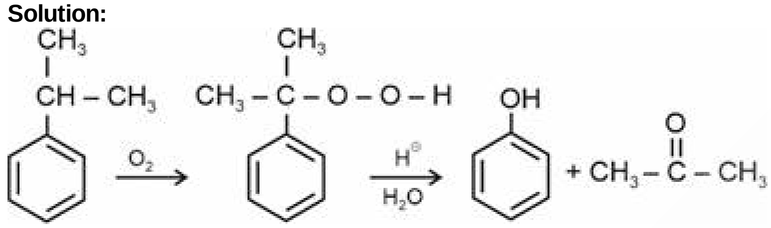
$$2\,CH_3\,CH_3 + 3O_2 \xrightarrow[\Delta]{(CH_3\,COO)_2\,Mn} 2\,CH_3\,COOH + 2H_2O$$

**(72)** **Answer :** (4)

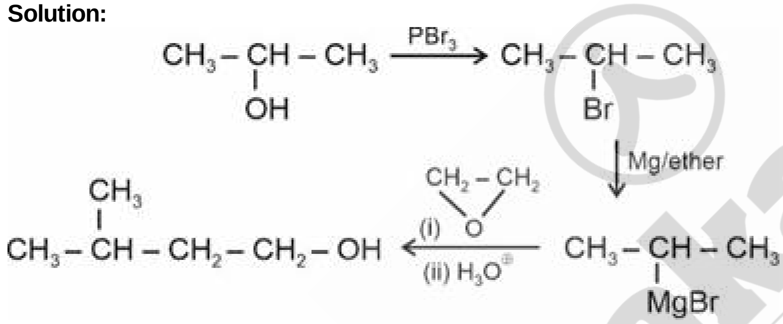
**Solution:**




**(73)** **Answer :** (3)
**Solution:**



**(74)** **Answer :** (1)
**Solution:**



**(75)** **Answer :** (3)
**Solution:**
Boiling point: $H_2O > H_2Te > H_2Se > H_2S$

**(76)** **Answer :** (4)
**Solution:**
$\mathrm{Cl_2 + \underset{\text{(cold \& dil.)}}{2\,NaOH} \rightarrow NaCl + H_2O + NaOCl}$

$\mathrm{3\,Cl_2 + \underset{\text{(hot \& conc.)}}{6\,NaOH} \rightarrow 5NaCl + NaClO_3 + 3H_2O}$

**(77)** **Answer :** (2)
**Solution:**
$P_4 + 3NaOH + 3H_2O \rightarrow PH_3 + 3NaH_2PO_2$

**(78)** **Answer :** (2)
**Solution:**
Ionisation enthalpy of Na 496 kJ mol$^{-1}$ and Li = 520 kJ mol$^{-1}$
Electron gain enthalpy of O = – 141 kJ mol$^{-1}$ and S = – 200 kJ mol$^{-1}$

**(79)** **Answer :** (3)
**Solution:**
For isoelectronic species; as the charge increase; Zeff increases, hence size decreases.

**(80)** **Answer :** (2)
**Solution:**
I. $Cu^{2+} + 2e^- \rightarrow Cu$; E° = 0.34 V

$\Delta G_1^\circ = -nFE^\circ$
$= -2F(0.34)$
II. $Cu^{2+} + e^- \rightarrow Cu^+$; E° = 0.16 V
$\Delta G_2^\circ = -nFE^\circ$
$\Delta G_2^\circ = -1\ F(-0.16)$
Subtracting (II) from(I)
$Cu^+ + e^- \rightarrow Cu$
$-FE^0 = -2F(0.34) + 0.16\ F$
E° = 0.52 V

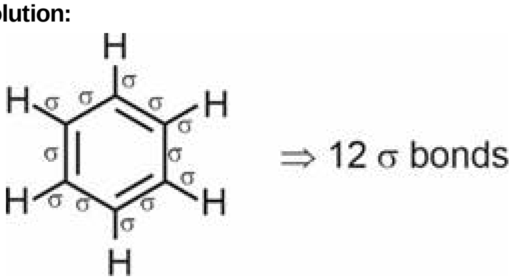
**(81) Answer :** (3)
**Solution:**
$\Lambda_m^0(CH_3\,COOH) = \Lambda_m^0(CH_3\,COOK) + \Lambda_m^0(HCl)$
$-\Lambda_m^0(KCl)$
= y + x – z

**(82) Answer :** (3)
**Solution:**
$pH = \frac{1}{2}(pkw + pKa + logc)$
$= \frac{1}{2}\left(14 + 4.2 + \log\left(\frac{2.88}{1000}\frac{\times 1000}{\times 144}\right)\right)$
= 8.25

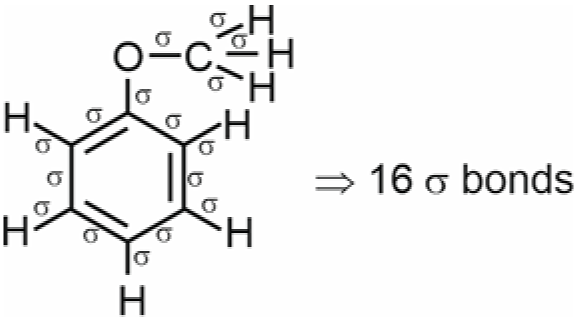
**(83) Answer :** (3)
**Solution:**
$AgCl\left(s\right) \rightleftharpoons \underset{S}{Ag^+}(aq) + \underset{S+0.2}{Cl^-}(aq)$
$k_{sp} = S(S + 0.2)$
$1.6 \times 10^{-10} = S(S + 0.2)$
$\Rightarrow 1.6 \times 10^{-10} = S(0.2)$ $\{\because (S + 0.2) \simeq 0.2\}$
$\Rightarrow S = \frac{1.6\times10^{-10}}{0.2}$
$\Rightarrow S = 8 \times 10^{-10}$

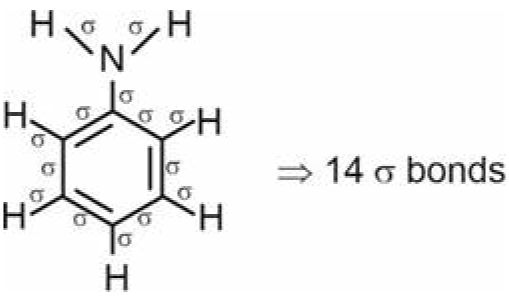
**(84) Answer :** (4)
**Solution:**

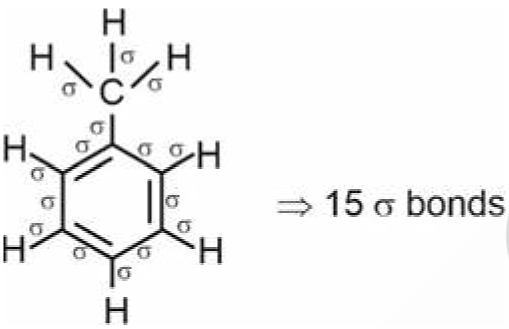
| Compound | Oxidation state of sulphur |
|---|---|
| $S_8$ | 0 |
| $SO_2$ | +4 |
| $H_2SO_4$ | +6 |
| $H_2S$ | –2 |

**(85) Answer :** (2)
**Solution:**
a. 

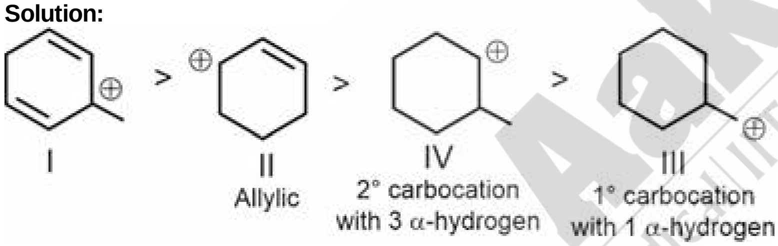

b.



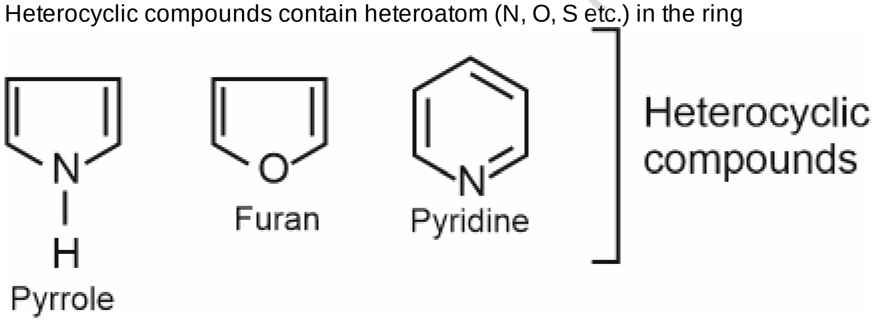

c. 

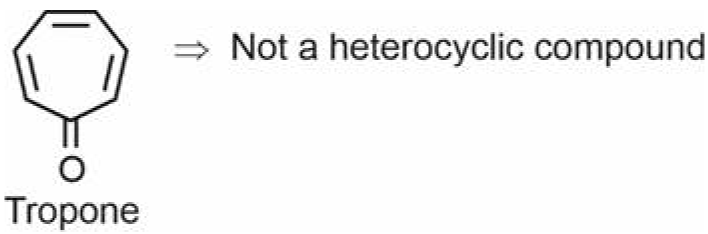

d. 


**(86)** **Answer :** (2)
**Solution:**




**(87)** **Answer :** (3)
**Solution:**
Heterocyclic compounds contain heteroatom (N, O, S etc.) in the ring






**(88)** **Answer :** (4)
**Solution:**
Six large chloride ions cannot be accommodated around $Si^{4+}$ due to limitation of its size and also, interaction between lone pair of chloride ion and $Si^{4+}$ is not very strong.

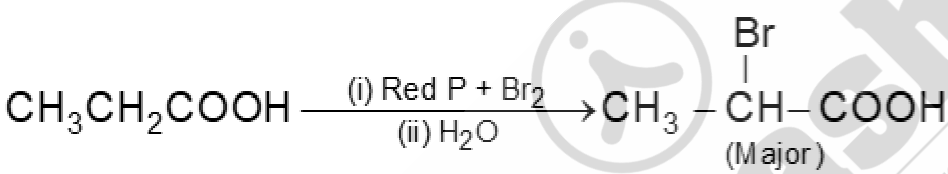
**(89)** **Answer :** (2)
**Solution:**
$\frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right)$

Or, $\frac{1}{\lambda} = R_H \left(\frac{1}{2^2} - \frac{1}{3^2}\right)$

Or, $\frac{1}{\lambda} = R_H \left(\frac{9-4}{36}\right)$

Or, $\frac{1}{\lambda} = \frac{5}{36} R_H$

**(90)** **Answer :** (2)
**Solution:**






**BIOLOGY**

**(91)** **Answer :** (1)
**Solution:**
The main plant body of the bryophytes is haploid. It produces gametes, hence is called a gametophyte.

**(92)** **Answer :** (3)
**Solution:**
In majority of the pteridophytes all, the spores are of similar kinds; such plants are called homosporous.

**(93)** **Answer :** (2)
**Solution:**
Cyclosporin A is used as an immunosuppressive agent in organ-transplant patients, which is produced by the fungus *Trichoderma polysporum.*

**(94)** **Answer :** (4)
**Solution:**
In competition, both the organisms get affected negatively (-,-).

**(95)** **Answer :** (2)
**Solution:**
In general, loss of biodiversity in a region may lead to decline in plant production, lowered resistance to environmental perturbations such as drought and increased variability in plant productivity, water use; pest and disease cycles.

**(96)** **Answer :** (1)
**Solution:**
The Nile perch introduced into Lake Victoria in east Africa led eventually to the extinction of an ecologically unique assemblage of more than 200 species of cichlid fishes in the lake.

**(97)** **Answer :** (4)

**Solution:**
In red algae, food is stored as floridean starch, which is very similar to amylopectin and glycogen in structure.

**(98) Answer :** (2)
**Solution:**
The primary effluent is passed into large aeration tanks where it is constantly agitated mechanically and air is pumped into it.

**(99) Answer :** (2)
**Solution:**
All birds and mammals and a very few lower vertebrate and invertebrate species, are able to maintain homeostasis by physiological as well as behavioural means.

**(100) Answer :** (2)
**Solution:**
DNA ligase joins the discontinuously synthesised fragments during replication.

**(101) Answer :** (1)
**Solution:**
RNA polymerase II transcribes hnRNA and RNA polymerase I transcribes rRNAs. Ribozyme contains 23S rRNA in bacteria.

**(102) Answer :** (2)
**Solution:**
Gymnosperms lack vessels in their xylem

**(103) Answer :** (1)
**Solution:**
Bast fibres - Generally absent in primary phloem
Companion cells - Maintain pressure gradient in sieve tubes
Casparian strips – Make endodermis impervious to water
Subsidiary cells - One of the components of stomatal apparatus

**(104) Answer :** (4)
**Solution:**
Satellite DNA does not normally code for proteins. It forms a larger portion of human genome. It sheds light on chromosome structure and dynamics.

**(105) Answer :** (1)
**Solution:**
Free 2'-OH of RNA makes it more labile and easily degradable as compared to DNA.

**(106) Answer :** (1)
**Solution:**
A unique vascular connection exists between the digestive tract and liver called hepatic portal system. Special venous connection between kidney and lower parts of body are present in frogs. It is called renal portal system.

**(107) Answer :** (2)
**Solution:**
The essential chemical components of many co-enzymes are vitamins, *e.g.*, co-enzymes nicotinamide adenine dinucleotide (NAD) and NADP contain the vitamin niacin. Co-enzymes are also organic compounds but their association with the apoenzyme is only transient, usually occurring during the course of catalysis.

**(108) Answer :** (4)
**Solution:**

| **Component** | **% of the total cellular mass** |
|---|---|
| Water | 70-90 |
| Proteins | 10-15 |
| Carbohydrates | 3 |
| Lipids | 2 |
| Nucleic acids | 5-7 |
| Ions | 1 |

Chemical compounds found in living organisms are of two types. One, those which have molecular weights less than one thousand dalton and are usually referred to as micromolecules or simply biomicromolecules while those which are found in the acid-insoluble fraction are called macromolecules or biomacromolecules. Lipids are not strictly macromolecules.

**(109) Answer :** (1)

**Solution:**
Adult human haemoglobin consists of 4 subunits. Two of these are identical to each other. Hence, two subunits of α type and two subunits of β type together constitute the human haemoglobin (Hb). They exhibit the quaternary structure of protein.

**(110) Answer :** (4)
**Solution:**

| | |
|---|---|
| Pigments | Carotenoids, Anthocyanins, *etc.* |
| Alkaloids | Morphine, Codeine, *etc.* |
| Terpenoides | Monoterpenes, Diterpenes, *etc.* |
| Essential oils | Lemon grass oil, *etc.* |
| Toxins | Abrin, Ricin |
| Lectins | Concanavalin A |
| Drugs | Vinblastin, curcumin, *etc.* |
| Polymeric substances | Rubber, gums, cellulose |

**(111) Answer :** (3)
**Solution:**
Exocrine glands secrete mucus, saliva, earwax, oil, milk, digestive enzymes and other cell products. These products are released through ducts or tubes. In contrast, endocrine glands do not have ducts. Their products called hormones are secreted directly into the fluid bathing the gland.

**(112) Answer :** (1)
**Solution:**
The nerve fibres of the PNS are of two types:
(a) Afferent fibres
(b) Efferent fibres
The afferent nerve fibres transmit impulses from tissues/organs to the CNS and the efferent fibres transmit regulatory impulses from the CNS to the concerned peripheral tissues/organs.
Pons consists of fibre tracts that interconnect different regions of the brain.

**(113) Answer :** (3)
**Solution:**
If any protein encoding gene is expressed in a heterologous host, it is called a recombinant protein. Transformation is a procedure through which a piece of DNA is introduced in a host bacterium.

**(114) Answer :** (2)
**Solution:**
*Agrobacterium tumefaciens*, a pathogen of several dicot plants, is able to deliver a piece of DNA known as 'T-DNA' to transform normal plant cells into a tumor and direct these tumor cells to produce the chemicals required by the pathogen. Similarly, retroviruses in animals have the ability to transform normal cells into cancerous cells.
The tumor inducing (Ti) plasmid of *Agrobacterium* has been modified into a cloning vector which is no more pathogenic to the plants but is still able to use the mechanisms to deliver genes of our interest into a variety of plants.

**(115) Answer :** (4)
**Solution:**
After completion of the biosynthetic stage, the product has to be subjected through a series of processes before it is ready for marketing as a finished product. The processes include separation and purification, which are collectively referred to as downstream processing. The product has to be formulated with suitable preservatives. Such formulation has to undergo thorough clinical trials as in case of drugs. Strict quality control testing for each product is also required.

**(116) Answer :** (3)
**Solution:**
Sclereids are present in the leaves of tea and seed coats of pea and beans.

**(117) Answer :** (3)
**Solution:**
In prokaryotes, DNA is negatively charged, contains 5-methyl uracil and packaged in the form of large loops held by basic proteins called polyamines.

**(118) Answer :** (4)
**Solution:**
Oxidative phosphorylation is a function of mitochondria. Mitochondrion is not a part of the endomembrane system.

**(119) Answer :** (1)
**Solution:**
(i) Secondary constriction is non staining, not the satellite.

(ii) Centriole has peripheral triplets.

**(120) Answer :** (4)
**Solution:**
Most of the organelles duplicate during the $G_1$ stage of cell cycle.

**(121) Answer :** (3)
**Solution:**
Crossing over takes place in the pachytene stage of cell cycle.

**(122) Answer :** (2)
**Solution:**
Interphase is also called as the resting phase.
ATP Synthesis occurs during metabolic pathways also.
Karyokinesis is not associated with the resting phase and takes place in M phase.

**(123) Answer :** (4)
**Solution:**
Ribosomes have RNA + protein

**(124) Answer :** (3)
**Solution:**
Secondary consumers are also called as primary carnivores and occupy the third trophic level.

**(125) Answer :** (2)
**Solution:**
ATP molecules are obtained when NADH + $H^+$ and $FADH_2$ are oxidised via ETS.

| | | |
|---|---|---|
| 1Pyruvic acid → | Link reaction →<br>1NADH | 1Krebs cycle<br>3NADH<br>1 $FADH_2$ |

4NADH + $H^+$ × 3 = 12
**1**$FADH_2$ × 2 = 2
= 14 ATP

**(126) Answer :** (3)
**Solution:**
Non-cyclic photophosphorylation results in the synthesis of $O_2$, ATP and NADPH.
Stroma lamellae lack PS II and NADP reductase enzyme.

**(127) Answer :** (1)
**Solution:**
Formation of two molecules of 3-PGA during Calvin cycle is due to carboxylase activity of RuBisCO.

**(128) Answer :** (1)
**Solution:**
Complete oxidation of one $FADH_2$ forms 2 ATP molecules and hence, two $FADH_2$ form 4 ATP molecules.

**(129) Answer :** (3)
**Solution:**
Miller et al. (1955) identified and crystallised the cytokinesis promoting active substance, termed as Kinetin.

**(130) Answer :** (3)
**Solution:**
Cell wall of the meristematic cells is primary in nature. During the elongation phase, cell wall of the cells start depositing new materials resulting in cell enlargement.

**(131) Answer :** (3)
**Solution:**
Recombinant DNA can be forced into competent cells by incubating the cells with recombinant DNA on ice, followed by placing them briefly at 42°C (heat shock), and then putting them back on ice. This enables the bacteria to take up the recombinant DNA.

**(132) Answer :** (4)
**Solution:**
The PNS is divided into two divisions. They are
(a) Somatic neural system and
(b) Autonomic neural system

The somatic neural system relays impulses from the CNS to skeletal muscles (Biceps) while the autonomic neural system transmits impulses from the CNS to the involuntary organs and smooth muscles of the body (Gall bladder, aorta and small intestine).

**(133) Answer :** (4)
**Solution:**
Traditional hybridisation procedures used in plant and animal breeding, very often lead to inclusion and multiplication of undesirable genes along with the desired genes.
The most commonly used matrix is agarose which is a natural polymer extracted from sea weeds.
The normal *E. coli* do not carry resistance against any antibiotics.

**(134) Answer :** (4)
**Solution:**

| | |
|---|---|
| Pivot joint | Between atlas and axis |
| Saddle joint | Between carpal and metacarpal of thumb |
| Ball and socket joint | Between humerus and pectoral girdle |
| Fibrous joint | In between bones of the cranium |

**(135) Answer :** (4)
**Solution:**
All chordates do not have a closed circulatory system (*e.g.* urochordates). All vertebrates have a closed circulatory system.
In fishes, heart pumps out deoxygenated blood which is oxygenated by gills.

**(136) Answer :** (2)
**Solution:**
Hyoid bone is present at the base of the buccal cavity. Temporal bone is present at the lateral part of the cranium.

**(137) Answer :** (4)
**Solution:**
In *Paramoecium*, cilia help in the movement of food through cytopharynx and in locomotion as well. *Hydra* uses its tentacles for locomotion. Amoeboid movement is shown by macrophages.

**(138) Answer :** (1)
**Solution:**
The intercellular material of cartilage is pliable because its ground substance is rich in chondroitin salts. Bones have a hard and non-pliable ground substance rich in calcium salts and collagen fibres which give bone its strength.

**(139) Answer :** (4)
**Solution:**
WBCs are nucleated and are relatively lesser in number which averages 6000-8000 $mm^{-3}$ of blood. They are nucleated.
Neutrophils are the most abundant cells (60-65 per cent) of the total WBCs.

**(140) Answer :** (2)
**Solution:**
Penis is the male external genitalia. It is made up of special tissue that helps in erection of the penis to facilitate insemination. Clitoris also has erectile tissues but does not facilitate insemination.

**(141) Answer :** (3)
**Solution:**
The vasa efferentia leave testis and open into epididymis located along the posterior surface of each testis. Rete testis is present in mediastinum of the testis which connects seminiferous tubules with vasa efferentia.

**(142) Answer :** (1)
**Solution:**
In cockroaches, simple eye (ocellus), anal cerci, antennae, *etc.*, act as sensory structures.
Pronotum is exoskeleton of prothorax of cockroach.

**(143) Answer :** (3)
**Solution:**
The male sex accessory ducts include rete testis, vasa efferentia, epididymis and vas deferens. Seminal vesicle is a male sex accessory gland. Penis is external genitalia of males.

**(144) Answer :** (4)
**Solution:**
In frogs, cutaneous respiration is a common mode of respiration when it is found on land as well as in water.

**(145) Answer :** (4)
**Solution:**

The process of delivery of baby is called parturition. Childbirth/ parturition is induced by a complex neuroendocrine mechanism. Soon after the infant is delivered, the placenta is also expelled out of the uterus.

**(146) Answer :** (3)
**Solution:**
Adrenal gland is composed of two types of tissues. The centrally located tissue is called adrenal medulla and outside of it lies the adrenal cortex. Adrenal medulla secretes two hormones called adrenaline or epinephrine and noradrenaline or norepinephrine.
Adrenal cortex secretes: Mineralocorticoids (*e.g* Aldosterone), glucocorticoid (*e.g* Cortisol) and androgenic steroids.

**(147) Answer :** (2)
**Solution:**
MALT(Mucosa-associated lymphoid tissue) is located within the lining of major tracts (respiratory, digestive and urogenital tracts).
It is a secondary lymphoid organ.

**(148) Answer :** (4)
**Solution:**
There is always a time-lag between the infection and appearance of AIDS symptoms. This period may vary from a few months to many years (usually 5-10 years).

**(149) Answer :** (2)
**Solution:**
Aldosterone acts mainly at distal tubules and stimulates reabsorption of $Na^+$ and water and excretion of $K^+$ and phosphate ions. PTH increases the $Ca^{2+}$ levels in the blood and stimulates reabsorption of $Ca^{2+}$ by the renal tubules.
TCT plays a significant role in maintaining the calcium balance in the body.
Oxytocin acts on smooth muscles of the body and stimulates their contraction.

**(150) Answer :** (4)
**Solution:**
Vector borne diseases *i.e.*, the diseases which are transmitted through insect vectors can be controlled by eliminating the vectors and their breeding places.
Introducing fishes like *Gambusia* in ponds that feeds on mosquito larvae can be helpful in eliminating such diseases. For example: Malaria, filariasis, chikungunya, dengue.

**(151) Answer :** (2)
**Solution:**
Lemur is a placental mammal which shows convergent evolution with spotted cuscus.

**(152) Answer :** (2)
**Solution:**
When more than one adaptive radiations appeared to have occurred in an isolated geographical area representing different habitats, one can call this convergent evolution.

**(153) Answer :** (4)
**Solution:**
Animals belonging to class Cyclostomata are ectoparasites on some fishes. They are marine but migrate to fresh water for spawning. Their larvae after metamorphosis return to the ocean. Cyclostomes are vertebrates.

**(154) Answer :** (4)
**Solution:**
The efferent arteriole emerging from the glomerulus forms a fine capillary network around the renal tubule called the peritubular capillaries.

**(155) Answer :** (3)
**Solution:**

| | |
|---|---|
| Renal corpuscle/Malpighian body | Glomerulus along with Bowman’s capsule |
| Bowman's capsule | Double walled cup-like structure |
| Renal pelvis | A broad funnel shaped space inner to the hilum |

**(156) Answer :** (1)
**Solution:**
When self- pollination occurs in the bud stage before the opening of flowers, it is called bud pollination. E.g. Pea, Rice

**(157) Answer :** (2)
**Solution:**
Pollination does not guarantee the transfer of the right type of pollen on the stigma of a particular plant species.

**(158) Answer :** (4)
**Solution:**
A-Antipodals, B-Polar nuclei, C-Synergids, D-Egg

**(159) Answer :** (1)
**Solution:**
In a typical test cross, an organism showing dominant phenotype is crossed with the recessive parent instead of self-crossing.

**(160) Answer :** (4)
**Solution:**
Klinefelter's male has tall stature.

**(161) Answer :** (2)
**Solution:**
Round yellow and wrinkled green seeds were obtained in the ratio of 9 : 1.

**(162) Answer :** (2)
**Solution:**
Colour blindness is an X-linked recessive disorder and in females, colour blindness appears only when both the sex chromosomes carry the recessive gene ($X^C X^C$) for which, her mother should atleast be the carrier of the disease and father should be colour blind.

**(163) Answer :** (3)
**Solution:**
Based on characteristics, living organisms can be classified into different taxa. This process of classification is taxonomy.

**(164) Answer :** (4)
**Solution:**
The cell membrane contains branched chain lipids which decreases membrane fluidity in archaebacteria.

**(165) Answer :** (4)
**Solution:**
Kingdom fungi is classified on the basis of the morphology of mycelium, mode of spore formation and fruiting bodies.

**(166) Answer :** (1)
**Solution:**
The floral formula of Fabaceae family is

$\% \ \text{⚥} \ K_{(5)} \ C_{1+2+(2)} \ A_{(9)+1} \ \underline{G}_1$

.

**(167) Answer :** (4)
**Solution:**
The hilum is a scar on the seed coat through which the developing seeds were attached to the fruit.

**(168) Answer :** (3)
**Solution:**
$K_{(n)}$ – Gamosepalous condition

**(169) Answer :** (2)
**Solution:**
Viruses are obligate parasites and no virus contains both RNA and DNA. Lichens are very good pollution indicators as they do not grow in the polluted areas.

**(170) Answer :** (2)
**Solution:**
Upright pyramid of number is observed when the number of producers is maximum.
Tree ecosystem will exhibit spindle shaped pyramid according to the given data.

**(171) Answer :** (4)
**Solution:**
Branching descent and natural selection are two key concepts of Darwinian theory of evolution. The concept of mutation and saltation was given by Hugo deVries.

**(172) Answer :** (3)
**Solution:**
*Adamsia* (Sea anemone) are radially symmetrical coelenterates. They have a polyp body form which is sessile and cylindrical. Digestion is both extracellular and intracellular.

**(173) Answer :** (2)
**Solution:**
First clinical gene therapy = 1990
Eli Lilly prepared humulin = 1983
Rosie produced human protein –
enriched milk = 1997
Enzymes responsible for restricting the growth of bacteriophage in *E. coli* was discovered in 1963

**(174) Answer :** (4)
**Solution:**
Patents should satisfy three criteria:
Novelty, non-obviousness and utility.

**(175) Answer :** (4)
**Solution:**
*Spongilla* is a fresh water sponge and it contains flagellated choanocytes.
*Euspongia* is found in marine water.
*Meandrina* and *Gorgonia* are coelenterates.

**(176) Answer :** (4)
**Solution:**
The intra uterine devices are presently available as the non-medicated IUDs (*e.g.*, Lippes loop), copper releasing IUDs (CuT, Cu7, Multiload-375) and the hormone releasing IUDs (Progestasert, LNG-20). Diaphragms, cervical caps and vaults are barriers made of rubber that are inserted into the female reproductive tract to cover the cervix during coitus.

**(177) Answer :** (3)
**Solution:**
Infertility cases either due to inability of the male partner to inseminate the female or due to very low sperm counts in the ejaculates could be corrected by artificial insemination (AI) technique. In this technique, the semen collected either from the husband or a healthy donor is artificially introduced either into the vagina or into the uterus.
(IUI - Intra-Uterine Insemination) of the female.

**(178) Answer :** (3)
**Solution:**
In amniocentesis, some of the amniotic fluid of the developing foetus is taken to analyse the foetal cells and dissolved substances. This procedure is used to test for the presence of certain genetic disorders such as, down syndrome, haemophilia, sickle-cell anemia, *etc*., and to determine the survivability of the foetus.

**(179) Answer :** (4)
**Solution:**

| Respiratory Gas | Atmospheric Air | Alveoli | Blood (Deoxygenated) | Blood (Oxygenated) | Tissues |
|---|---|---|---|---|---|
| $O_2$ | 159 | 104 | 40 | 95 | 40 |
| $CO_2$ | 0.3 | 40 | 45 | 40 | 45 |

**(180) Answer :** (4)
**Solution:**
$O_2$ and $CO_2$ are exchanged in the alveoli by simple diffusion mainly based on partial pressure/concentration gradient.
Solubility of the gases as well as the thickness of the membranes involved in diffusion are also some important factors that can affect the rate of diffusion.